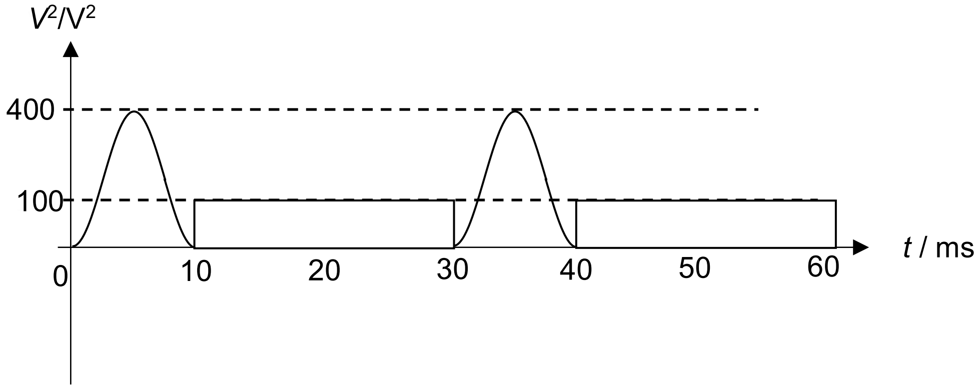
27 B Calculate r.m.s. voltage across the resistor. Then use P = $\frac{V_{rms}{}^2}{R}$ to find the mean power.



Mean of V^2 = $\frac{(200\times10)+(100\times20)}{30}$ = 4000/30 volts^2

Mean power = $\frac{V_{rms}{}^2}{R} = \frac{4000/30}{5.0} = 26.7\ \mathrm{W}$

28 B Originally, $eV = hf - \Phi$ ... $V = (hf - \Phi)/e$